

Deep Aman G

AI/ML Engineer

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SUMMARY

Passionate AI/ML Engineer with a strong foundation in Generative AI, NLP, and cloud-native development. I’ve applied these skills in impactful projects—such as building RAG systems and deploying intelligent agents during my time at Ford and Veninfos. I thrive on creating AI solutions that are practical, scalable, and meaningful in the real world. Skilled in leveraging AWS, Azure, and open-source tools to build end-to-end intelligent systems, I’m driven by curiosity and a desire to turn bold ideas into real-world impact.

EXPERIENCE

Ford Motor Company 2024-05 - Present
GenAI Developer

- Designed a RAG-based GenAI platform using LangChain and FAISS, enabling context-aware responses across compliance-critical documentation.
- Deployed a scalable semantic search pipeline on AWS EC2, leveraging vector indexing and embedding models for high-precision retrieval.
- Automated ingestion and processing of unstructured documents using AWS Lambda, with structured storage and retrieval from Amazon S3.
- Embedded Crew AI into internal systems to autonomously draft context-driven communications, reducing manual workload and improving response quality.

Veninfos 2022-06 - 2023-07
ML/NLP Developer

- Built and productionized an NLP workflow for sentiment classification using TF-IDF, Logistic Regression, and SVM, trained on e-commerce review datasets.
- Applied scalable preprocessing techniques including token normalization, lemmatization, and stop-word filtering for language modeling.
- Developed dynamic Tableau dashboards to present sentiment trends, user behavior insights, and category-based feedback patterns.
- Packaged and deployed the model as a Flask REST API on AWS EC2 to support real-time inference and integration into analytics workflows.

Smart Bridge collaboration with IBM 2022-02 - 2022-04
Developer Intern

- Designed a CNN using VGG16 architecture to detect leaf diseases in fruits and vegetables. Developed a user-friendly web interface for diagnosis.
- Designed a CNN using VGG16 architecture to detect leaf diseases in fruits and vegetables, achieving high accuracy.
- Trained the model on a dataset of over 20,000 images and deployed it as a Flask-based API.
- Developed a user-friendly web interface using HTML and CSS, enabling farmers to upload leaf images for diagnosis.
- Enhanced agricultural decision-making, reducing crop diseases and increasing yields.

CERTIFICATIONS

Azure AI Engineer Associate Microsoft Azure	AWS Machine learning Associate AWS	Generative AI: Prompt Engineering IBM, Coursera
Generative AI with Large Language Models , Specialization in GANs	AWS Cloud Fundamentals AWS and Coursera	Azure cloud Essentials Azure and Coursera

EDUCATION

University at Buffalo, The State University of New York (3.6/4.0 GPA) https://www.buffalo.edu/	2023-08 - 2024-12 Master of Science - Data Science
Gudlavaluru Engineering College (3.7/4.0 GPA)	2019-08 - 2023-08 Bachelor of Technology - Computer Science

SKILLS

Artificial Intelligence and Machine Learning

AI-driven systems, Distributed systems, Large-scale predictions, Decision Trees, Random Forest, Gradient Boosting (GBM), SVM, K-Means, KNN

Deep Learning

CNN, RNN, Transformers, Sequence-to-Sequence Models

Generative AI

RAG, Intelligent Agents, Chatbots, RLHF, PEFT, Automated Text Generation

Azure Skills

Azure Cognitive Search, Azure Open AI Service, Azure Cognitive Services, Computer Vision, Text Analytics, Language Understanding (LUIS), Document Intelligence

Distributed Systems and Databases

Relational Databases: MySQL, PostgreSQL, NoSQL: DynamoDB, CosmosDB

Programming and Development

Python, Java, Object-Oriented Design, Data Structures, Complexity Analysis

Cloud Computing and Infrastructure

Azure Functions, AWS EC2, Lambda, S3, RDS, Docker, Kubernetes, Git, CI/CD pipelines

Data Science and Visualization

NumPy, Pandas, Scikit-learn, Tableau, Power BI, Matplotlib, Seaborn

Natural Language Processing (NLP)

Tokenization, NER, Sentiment Analysis, Word Embeddings (Word2Vec, GloVe)

Responsible AI Practices

Security, Privacy, Accessibility, Microsoft Responsible AI

PUBLICATIONS

Performance Analysis of VGG Algorithms on Pneumonia Classification Scopus	Machine Learning for Predictive Analysis of Chronic Liver Disease Scopus
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